Handwritten Character and Word Recognition with a CNN–BiGRU–CTC Network: A Comparison with Classical and Pretrained Baselines

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***Abstract*—Handwriting recognition is difficult because of writer variability, connected strokes and visually similar characters. This paper builds a handwritten character and word recognizer around a convolutional recurrent neural network (CRNN) that combines a CNN feature extractor, a bidirectional gated recurrent unit (BiGRU) and connectionist temporal classification (CTC) decoding, and compares it with a classical HOG + linear SVM baseline, an SE-ResNet character classifier, and the pretrained TrOCR model. On the EMNIST ByClass test set (116,323 images), HOG + SVM reaches 78.83% accuracy and the SE-ResNet reaches 87.33% accuracy and 74.01% macro F1. The CRNN, trained on synthetic words stitched from EMNIST characters, reaches 17.78% character error rate (CER) on a fixed synthetic test set, but it transfers poorly to real handwriting. We report these results together with their limits.**

***Index Terms*—handwriting recognition, CRNN, GRU, CTC, EMNIST, HOG, SVM**

I. Introduction

Handwriting recognition must cope with large differences between writers, touching or cursive strokes, and confusable classes such as `O`/`0` and `I`/`l`/`1`. Segmenting a word into characters before classifying them fails when strokes connect. A CRNN avoids segmentation by reading the image as a sequence and decoding it with CTC.

The contributions of this work are:

- A single CRNN (CNN + BiGRU + CTC) that serves single-character, word and paragraph-line input in one web application.
- A seeded comparison of BiLSTM and BiGRU recurrent layers under identical data, data order and epochs.
- A baseline study on the same EMNIST splits: HOG + linear SVM and an SE-ResNet classifier, with a pretrained TrOCR model as a real-handwriting reference.
- An evaluation that keeps synthetic-data results separate from real-handwriting behaviour.

II. Related Work

EMNIST extends MNIST to letters [1]. Histograms of oriented gradients (HOG) [2] summarize local stroke directions and, with a linear classifier, form a standard pre-deep-learning baseline. Residual networks [3] and squeeze-and-excitation blocks [4] improve convolutional classifiers. CRNNs combine convolutional features with recurrent layers for sequence recognition [5], trained with CTC [6]. The GRU [7] is a lighter alternative to the LSTM. TrOCR [8] is a pretrained transformer line recognizer, and the IAM database [9] is the standard line-level benchmark.

III. Datasets

EMNIST ByClass has 62 classes (digits, uppercase and lowercase letters) as 28×28 grayscale images. The stored image orientation is corrected before use. The character classifiers are trained on the ByClass training split and evaluated on the official test split of 116,323 images.

Synthetic words. The CRNN is trained on random space-free character strings stitched from EMNIST *balanced* (47 classes) characters. Because they are built from isolated characters, they do not capture natural word-level handwriting. Balanced merges several upper- and lowercase letter pairs into one class, so the CRNN cannot distinguish case for those letters.

IV. Methodology

A. HOG + linear SVM

Each image is converted to a 324-dimensional HOG descriptor (9 orientations, 7×7-pixel cells, 2×2-cell blocks). A multiclass linear SVM with hinge loss is trained incrementally for five epochs on the ByClass training split.

B. SE-ResNet character classifier

The classifier uses residual blocks, squeeze-and-excitation channel attention and global average pooling. It is trained with AdamW, a OneCycleLR schedule and data augmentation.

C. CRNN

The CRNN has three parts: (i) a convolutional feature extractor that turns the image into a left-to-right sequence of feature columns; (ii) a bidirectional recurrent layer (GRU or LSTM) that adds context from both directions; and (iii) a linear output layer trained with CTC loss, which learns the alignment between feature columns and characters without character-level labels. At inference, greedy decoding collapses repeated symbols and removes blanks. Single characters are treated as one-character sequences, and text images are split into line crops before decoding.

TABLE I
ISOLATED-CHARACTER RESULTS ON THE EMNIST BYCLASS TEST SET (116,323 IMAGES).

Model	Accuracy (%)	Macro F1 (%)
HOG + linear SVM	78.83	—
SE-ResNet CharCNN	87.33	74.01

TABLE II
BILSTM-CTC VS BIGRU-CTC, 3-EPOCH CPU COMPARISON ON 512 SYNTHETIC TEST WORDS.

Model	CER (%)	WER (%)	Exact (%)
BiLSTM-CTC	100.00	100.00	0.00
BiGRU-CTC	97.31	100.00	0.00

D. TrOCR reference

Pretrained TrOCR (`base-handwritten`) is used only as a command-line reference for real handwriting. It is not trained or modified in this work.

V. EXPERIMENTAL SETUP

All CRNN experiments use seed 42 and run on a CPU. The application CRNN uses a bidirectional GRU and was trained for 10 epochs on 5,000 synthetic words per epoch; it is evaluated on 1,000 validation words and a separate fixed set of 512 synthetic test words. The BiLSTM/BiGRU comparison uses 1,024 training, 256 validation and 512 test words, with identical data order and epochs for both models.

Character classification is reported as accuracy and macro F1. Sequence recognition is reported as CER, word error rate (WER) and exact match. Each synthetic sample is one space-free string, so synthetic WER is effectively a whole-string error rate and is not comparable to sentence-level WER.

VI. RESULTS

A. Isolated-character classification

The CNN improves on the classical baseline by 8.5 accuracy points. The gap between accuracy and macro F1 is consistent with class imbalance and confusable pairs in ByClass.

B. CRNN on synthetic words

The application checkpoint reaches 39.9% exact match on 1,000 validation words and 17.78% CER, 58.59% WER and 41.41% exact match on the 512-word synthetic test set.

At this small budget neither model learned useful decoding, so Table II is an architecture smoke test rather than a competitive comparison.

C. Real handwriting (qualitative)

Table III shows outputs on two real samples. These are examples, not an accuracy measurement.

The 10-epoch application CRNN read `Dee` as `DEE` and a paragraph sample as `MMKM`. The case error on `Dee` is consistent with the balanced label set merging case for some letters; this explanation has not been tested.

TABLE III
REAL-IMAGE OUTPUTS (TWO EXAMPLES ONLY).

Image (ref.)	BiLSTM	BiGRU	Contour+CNN	TrOCR
`Dee`	(empty)	(empty)	`Dee`	`De e`
`B8Cd7`	(empty)	(empty)	`B8Cd7`	`B8cd7`

VII. DISCUSSION

The synthetic-data score does not transfer to real handwriting. Stitched EMNIST characters have uniform spacing and stroke width, which natural writing lacks, and the CRNN has never seen connected strokes. On the two isolated, clearly separated examples, contour segmentation with the CNN did better than the CRNN, which suggests that the CRNN's advantage appears only when characters touch, a case not measured here.

The CRNN and the SE-ResNet use different label sets (balanced vs ByClass), so their numbers should not be compared directly.

VIII. LIMITATIONS

Synthetic results do not measure general handwriting accuracy. The real-image evaluation has two samples. The evaluation covers no phone photos or punctuation. The BiLSTM/BiGRU comparison used a 3-epoch budget, and no confidence intervals or multiple seeds are reported.

IX. CONCLUSION AND FUTURE WORK

A CNN–BiGRU–CTC network recognizes characters and words without segmentation, but synthetic training limits it on real handwriting. The SE-ResNet beats a HOG + SVM baseline by 8.5 accuracy points on EMNIST ByClass. Future work includes training the CRNN on real line images such as IAM, using the ByClass label set, more varied synthetic data, and a larger real-image evaluation set.

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